A

Report for

TD619: Energy Policy and Planning

On

Residential Grid-Connected Rooftop PV Systems

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Summary

The deployment, regulations, stakeholder viewpoints, and financial feasibility of residential grid-connected rooftop photovoltaic (GRPV) systems are all examined in detail in this research. It looks at global patterns in China, India, Japan, and the United States, highlighting important facilitators such as declining solar prices, legislative incentives, and growing environmental awareness. Even though India has 106.65 GW of total solar capacity, only 3.2 GW of that is used for residential GRPV. To encourage adoption, the government offers subsidies through programs like PM Surya Ghar Yojana (PMSGY). Two notable examples of progressive state policy are Telangana's net metering changes and Kerala's SOURA program. The study also assesses Kerala's billing practices, which have improved rooftop solar's feasibility. These practices include net metering, time-of-day changes, and payment techniques. Solar installers, technicians, MSEDCL engineers, and customers from urban, semi-urban, and rural regions provided input as stakeholders. Roof shadowing, structural problems, and subsidy knowledge were among the installation concerns mentioned by the installers. The long-term financial and ecological advantages of rooftop solar were highlighted by engineers. According to consumer surveys, after installation, the use of electric equipment increased and electricity expenses decreased by 80–100%. Nonetheless, issues with inconsistent rules, poor service in rural regions, restricted access to skilled technicians, and delays in subsidy payouts persisted. Due to the availability of free electricity under state assistance programs and a lack of maintenance, users of off-grid devices reported abandoning them. Assuming ₹18,000 per 500W panel and an IRR of 6%, the report models system configurations and assesses financial viability using HOMER software. In order to represent common grid-connected configurations, battery storage was not included. The results demonstrated that GRPV systems may provide short payback periods and were economically feasible, particularly when subsidised. The research ends with suggestions for bettering after-sales support, awareness campaigns, and expedited regulatory processes. Unlocking India's full home rooftop solar potential and advancing the nation's larger clean energy objectives need coordinating federal and state policies, guaranteeing financial accessibility, and cultivating customer trust.

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1. Introduction

To fight against increasing global warming, depleting fossil fuel reserves, and the rising demand of electricity, the transition to decentralized energy sources has become necessary. Among renewables, solar energy stands out as a clean and scalable solution. Grid Connected Rooftop Solar Photovoltaics (GRPV) represent an important component to enable consumers to become producers of electricity or “prosumers”. In addition to providing the chance to lower electricity costs and increase energy independence, GRPV systems also make a substantial contribution to lowering carbon emissions and easing the strain on the centralised grid infrastructure. India has been aggressively marketing rooftop solar systems through national and state-level efforts due to its enormous solar potential and high renewable energy targets. Nevertheless, the residential rooftop sector only makes up a minor portion of the total solar capacity, which exceeds 100 GW. Widespread adoption has been hampered by issues such low domestic electricity pricing, large upfront costs, a lack of customer knowledge, inconsistent policies, and opposition from distribution firms (DISCOMs). Meanwhile, creative policy frameworks—like Telangana's net metering laws and Kerala's SOURA scheme—showcase the revolutionary potential of thoughtfully planned government initiatives. These initiatives have had a noticeable effect on adoption, particularly when backed by financial incentives, streamlined processes, and strong post-purchase assistance.

2. GRPV Deployment in India and Other Countries

2.1. GRPV in USA

2.1.1 Current status of GRPV deployment [4]

As of 2024, the United States has seen significant growth in grid-connected rooftop photovoltaic (GRPV) installations. In 2010, residential solar capacity was 667 megawatts (MW), which surged to over 18,061 MW by 2020 which is a 27-fold increase. This growth is attributed to decreasing installation costs and supportive policies [1].

Interconnection standards, which define how distributed generation systems like rooftop solar connect to the grid, vary across states. Inconsistent or complex interconnection processes can pose challenges, potentially increasing costs and causing delays. Efforts are ongoing to streamline these procedures to facilitate easier integration of rooftop solar systems [2].

Community solar projects are also expanding, providing alternatives for those unable to install rooftop panels. As of late 2024, community solar had approximately 6.5 gigawatts of installed capacity in the U.S., with expectations to double by 2028 [3].

2.1.2 Potential for growth and expansion in these systems [5]

The National Renewable Energy Laboratory (NREL) estimates that approximately 3.3 million homes per year will be built or require roof replacements, representing a potential of roughly 30 gigawatts (GW) of solar capacity annually. Even if a small fraction of these new roofs incorporate solar installations, it could significantly impact U.S. solar power generation [6].

The U.S. Department of Energy's SunShot Initiative has been instrumental in reducing the cost of solar energy, making it more competitive with traditional energy sources. The initiative's efforts have led to significant decreases in the cost of solar electricity, contributing to increased adoption rates [7].

Challenges such as inconsistent interconnection standards across states can pose barriers to GRPV deployment. Complex interconnection processes may increase costs and cause delays, underscoring the need for streamlined procedures to facilitate easier integration of rooftop solar systems [8].

2.1.3 Key factors [5]

Enablers:

1. Declining Costs: The decreasing cost of PV modules and associated components has made rooftop solar installations more financially accessible for homeowners.
2. Policy Support: Federal and state incentives, such as tax credits and rebates, encourage homeowners to invest in solar energy systems.
3. Technological Advancements: Improvements in solar technology have enhanced system efficiency and reliability, making rooftop PV systems more appealing.
4. Environmental Awareness: Growing concern about environmental sustainability motivates individuals to adopt clean energy solutions like rooftop solar.

Barriers:

1. High Initial Costs: Despite declining prices, the upfront investment for purchasing and installing rooftop PV systems can still be prohibitive for many homeowners.
2. Regulatory Challenges: Inconsistent interconnection standards and permitting processes across different jurisdictions can complicate and delay installations.

3. Market Barriers: Limited access to financing options and a lack of consumer awareness can hinder the widespread adoption of rooftop PV systems.
4. Recycling and End-of-Life Management: The absence of comprehensive policies for recycling PV modules poses environmental concerns and may affect public perception of solar energy's sustainability.

2.2. GRPV in Japan

2.2.1 Current status of GRPV deployment

A total capacity of 84.9 GW has been deployed by 2022, with 1 GW in the residential sector. Japan targets to install 147 GW cumulative PV capacity to meet carbon reduction up to 50% by 2030. Japan introduced a comprehensive FiT scheme in July 2012 to promote RE diffusion [8].

2.2.2 Potential for growth and expansion in these systems

The Japanese environment and trade ministries have also published a target to achieve 108 GW of solar power capacity by 2030, up-scaling its previous 64 GW target by 1.7 times [9]. Currently, Japan is on track to reach 88 GW of solar capacity by 2030, implying that an additional 20 GW is needed to hit the 108 GW target [9]. Meeting these 2030 targets will require a near-doubling of Japan's total installed PV capacity from 2022 levels [10].
To achieve these ambitious goals, the government has outlined specific strategies:

1. Installing solar panels on 50% of central government and municipality buildings, which is expected to contribute 6 GW of capacity [9].
2. Adding another 10 GW from solar panels on corporate buildings and car parks [9].
3. Developing public land and promotion areas in at least 1,000 towns and cities, contributing an additional 4 GW to the goal [9].

Looking further ahead, according to BloombergNEF (BNEF), Japan needs to significantly increase its installed solar and wind power generation capacity by more than eight times – from 81 GW in 2021 to 689 GW in 2050 – if it is to reach its net zero targets. With such capacity, solar and wind generation would account for 79% of the electricity supply while nuclear would provide 11%.

2.2.3 Key Factors [11]

Enablers:

1. Supportive policy framework: Japan's renewable energy growth has been largely driven by supportive government policies. A comprehensive Feed-in Tariff (FiT) scheme

introduced in July 2012 significantly promoted renewable energy diffusion following the Fukushima nuclear disaster. This scheme fixed FiT rates for systems less than 10 kW for ten years and systems greater than 10 kW for 20 years. The policy landscape continues to evolve with the introduction of a Feed-in Premium (FiP) in April 2022, providing a more market-oriented approach to renewable energy support. This mechanism aims to ensure that solar producers are more responsive to market signals and integrate better into the overall energy market [11].

2. Regulatory Mandates: Tokyo is taking a pioneering step by mandating that all new homes in the city be built with rooftop solar panels starting in 2025. This mandate, the first of its kind in Japan, applies to roughly 50 large construction companies, which will be required to install solar arrays on homes with up to 2,000 square meters of floor space. Such regulatory requirements can significantly drive adoption in urban areas [12].
3. The government is funding innovative solutions through initiatives like the Green Innovation Fund with a total budget of JPY 123.5 billion. Additionally, ANRE is supporting the development of next-generation solar cell technology through the "Next Generation Solar Cell Development Project," costing JPY 64.8 billion [13].

Barriers:

1. Space Constraints: Japan has limited land area and densely populated cities. Many urban areas have small rooftops or complex building structures that make it difficult to install solar panels. The uneven availability of suitable roofs, especially in older buildings, reduces the potential for rooftop solar installations.
2. High Installation Costs: While the cost of solar panels has decreased globally, installation costs in Japan remain relatively high due to expensive labor, regulatory compliance and paperwork costs and specialized equipment for Japan's unique architectural designs.
3. Regulatory and Administrative Hurdles: Japan has strict building codes and regulations for safety and earthquake resistance, which can make rooftop solar installations more challenging. Complex approval processes and delays in connecting to the grid discourage homeowners and businesses from adopting solar energy.
4. Lack of Grid Capacity: The country's grid system faces regional fragmentation, with different frequency systems (50 Hz in eastern Japan and 60 Hz in western Japan). Limited grid capacity and lack of infrastructure to handle decentralized energy generation make it harder to integrate rooftop solar systems effectively.

5. Weather and Natural Disasters: Japan is prone to typhoons, earthquakes, and heavy snowfall, which can damage solar panels or make installations risky. Homeowners may be hesitant to invest in rooftop solar due to concerns about maintenance and reliability in such conditions.

2.3. GRPV in China

2.3.1 Current status of GRPV deployment

As of the end of 2023, China had installed approximately 220 GW of grid-connected rooftop solar photovoltaic (GRPV) capacity, making up around 43% of its total solar PV capacity of 617 GW [14]. This marks a rapid increase from 132 GW in 2021, indicating a 66% growth in just two years [15].

In 2022, China added 51 GW of rooftop solar PV capacity, accounting for 55% of all new solar PV additions that year [16]. A major driver of this growth was the "Whole County Rooftop Solar Pilot Program", launched by China's National Energy Administration (NEA) in 2021. Under this program, 676 counties were selected, with an estimated combined potential to add over 300 GW of rooftop PV capacity [17].

In China, "counties" are administrative divisions that fall below prefectural-level cities and above townships in the country's governance structure. They are equivalent to districts in other countries.

The distribution of GRPV installations across sectors as of 2023 is approximately:
Residential rooftops: ~40%
Commercial and industrial rooftops: ~55%
Public buildings and agricultural facilities: ~5% [18]

2.3.2 Potential for Growth and Expansion of GRPV in China

Market Potential:
China's GRPV market remains highly underexploited despite rapid growth. According to NEA and research agencies:

- The technical potential for rooftop PV in China is estimated at 3,000 GW (3 TW) [19].
- Only about 7% of this potential had been utilised by the end of 2023, with ~220 GW installed [20].

- Rural areas and industrial parks are expected to lead the next phase of expansion due to large roof availability and lower shading issues.

Forecasted Growth:

- The 14th Five-Year Plan (2021–2025) targets at least 50% of new building rooftops to be equipped with solar PV systems [21].
- BloombergNEF projects that China will add 70–90 GW of rooftop solar annually between 2024–2030, with a total installed rooftop PV capacity potentially reaching 700 GW by 2030 [22].
- The "Whole County Pilot Program" alone is expected to unlock over 300 GW of rooftop installations by 2025 [23].

2.3.3 Key Factors

Enablers:

Financial Incentives and Subsidies:

Although national subsidies for residential PV were phased out in 2021, several key financial enablers continue to support growth:

- Whole County Pilot Program: Offers 0.2–0.3 RMB/kWh in performance-based incentives at the local level, funded by provincial governments [22].
- Green Electricity Trading Markets: GRPV system owners can sell excess electricity at market-determined rates (often 0.35–0.45 RMB/kWh) via provincial energy exchange platforms [23].

Access to Green Finance:

- China's central bank has included GRPV projects in its Green Bond Endorsed Project Catalogue (2021 edition) [24].
- Specialised green loans from major banks (e.g., Bank of China, ICBC) offer:
    - Interest rates: ~3.5%–4.2%
    - Tenure: 8–10 years
    - Loan-to-value (LTV): Up to 85% for GRPV installations [25]

Policy and Regulatory Support:

- 14th Five-Year Plan (2021–2025): Mandates new buildings to integrate rooftop PV, with implementation targets across provinces [27].
- Carbon Neutrality Roadmap (to 2060): Encourages distributed PV to reduce grid burden and support energy decentralisation [28].

- Local mandates in provinces like Hebei, Jiangsu, and Zhejiang require all new industrial buildings to have PV-ready roofs [29].

Barriers:

Structural and Rooftop Suitability Issues:

- Despite large theoretical potential, only ~30% of rooftops in urban residential buildings are structurally suitable for PV installations [30].
- Many apartment buildings lack shared ownership agreements, making permission from all residents mandatory, which prevents installation [31].
- In rural areas, older rooftops often require structural reinforcement, adding 5–10% extra cost to installation [32].

High Upfront Costs for Households:

- While module prices have fallen, average residential GRPV systems (5–10 kW) still cost 25,000–50,000 RMB, which is significant for low-income rural households [33].
- The phase-out of central government subsidies in 2021 has increased the financial burden on homeowners, especially in provinces without strong local incentives [34].

Declining Tariff Rates and Returns:

- The end of the national feed-in tariff (FiT) program led to a drop in electricity buyback rates. In many regions:
  - FiT was 0.42 RMB/kWh (before 2021)
  - Post-policy rates dropped to 0.30–0.35 RMB/kWh, reducing Internal Rate of Return (IRR) [35].
- For residential GRPV systems, the average payback period increased from 6–8 years (with FiT) to 9–12 years (without FiT).

2.4. GRPV in India

2.4.1 Current status of GRPV deployment

As of March 2025, India's total installed solar power capacity stands at 106.65 GW[36], with Grid-Connected Rooftop Photovoltaic (GRPV) systems contributing 17.02 GW to this total. Within the GRPV segment, the residential sector accounts for 3.2 GW as of March 2024—approximately 27% of the total rooftop solar capacity[37]. Notably, around 75% of this residential capacity is concentrated in Gujarat, highlighting the state's leadership in household solar adoption. The data suggests that the majority of rooftop solar installations are driven by the industrial and commercial sectors, underlining their significant role in India's decentralized renewable energy landscape.

2.4.2 Potential for growth and expansion in these systems [37]

1. Enormous Technical Potential (637 GW)

- India has a technical rooftop solar potential of 637 GW, assuming all suitable rooftops are utilized.
- However, actual usable capacity reduces to one-fifth when limited by household electricity consumption due to generally low per square foot electricity usage in most states.

2. Economic Potential Limited by Low Tariffs (102–118 GW)

- Only 118 GW is viable when RTS system sizes are matched to household electricity demand.
- Drops further to ~102 GW when economic viability (NPV > 0) is considered.
- Low residential electricity tariffs make rooftop systems financially unappealing for many, even if technically feasible.
- This causes a 16 GW drop from technical to economic potential.

3. Market Potential Severely Limited (~11 GW)

1. Market potential is only ~11 GW, considering consumer willingness to pay and short payback expectations.
2. Consumers expect payback within 5 years, but:
    a. High system costs
    b. Low tariffs
    c. Poor awareness of benefits makes this difficult to achieve.

2.4.3 Key Factors

Enablers:

1. Favorable Solar Conditions: India receives an average solar insolation of 4-7 kWh/m²/day across most regions, providing ideal conditions for solar energy.

2. Cost Competitiveness: Declining prices of solar PV panels and associated equipment make rooftop PV systems more economically viable.
3. Energy Demand and Urbanization: Rapid urbanization and increasing residential electricity demand highlight the potential of rooftop solar power as an alternative to grid electricity.
4. Electricity tariffs: Rising electricity tariffs in urban areas, driven by escalating demand—are reducing the payback period of GRPV systems, thereby accelerating their economic viability and adoption.
5. Decentralized Power Generation: Rooftop PV reduces dependency on centralized power plants, minimizing transmission losses and enhancing energy security.
6. Government Initiatives:
   - With increased capital subsidy under schemes such as PMSGY and improved consumer awareness and understanding of rooftop solar technology, the outlook for India's residential rooftop solar market appears highly promising.
   - Higher rooftop space availability in non-metropolitan areas, and enhanced access to viable financing options, market demand will be evenly spread across tier-1 urban areas and tier-2,3/rural regions. The average project size is likely to drop below 3kWp since homes in rural areas have lower power requirements, leading to smaller rooftop solar plant capacity.

Barriers:

1. Low Residential electricity tariffs: Subsidized electricity tariffs—predominantly in Tier 2/3 cities and rural areas—diminish the economic appeal of rooftop solar installations. For low-consumption households, the resulting savings are often marginal, further eroding financial viability and disincentivizing adoption.
2. High upfront costs: Installation costs remain a significant hurdle, especially for low-income groups.
3. Awareness & Trust Issues: Limited understanding of benefits and subsidies and concerns about vendor reliability and service quality.

4. Maintenance and after sales service: This issue is particularly pronounced in remote and semi-urban areas, where access to qualified technicians and spare parts may be limited. Inadequate servicing can lead to reduced system efficiency, increased downtime, and lower overall returns on investment.

5. Policy and Regulatory Barriers: Electricity is a subject under the jurisdiction of individual states, rather than the central government. As a result, there is significant inconsistency in net metering policies across the country. While some states have adopted progressive frameworks to encourage rooftop solar adoption, others have imposed restrictive caps, complex procedures, or frequent policy changes, creating uncertainty for consumers and investors.
    - Delays in approvals and resistance from distribution companies (DISCOMs), concerned about potential revenue losses, further hamper the deployment process.
6. Limited Access to Financing: Difficulty in accessing affordable and timely financing, particularly for residential and small commercial consumers. Many potential adopters face challenges in securing loans, owing to limited credit history, collateral requirements, or lack of awareness about available schemes.

    - There is a shortage of standardized and tailor-made financial products for rooftop solar, which further discourages investment.

3. Central and State Government Policies and Regulations in India

3.1. Policies and Regulations in India

1. **Grid Connected Rooftop Solar Programme [38]:**

   The objective of the programme is to achieve a cumulative installed capacity of 4 GW from Grid Connected Rooftop Solar (RTS) projects. The program is structured in two components to provide both financial assistance and incentives for different stakeholders.

   Component A(CFA to residential sector): Financial aid is provided to residential consumers who install solar roof panels. The government aims to support the installation of rooftop solar systems across residential homes with a combined capacity of 4 GW.

   Table 1: Subsidies given by central government

Central Financial Assistance (CFA)/ Central Government Subsidy for rooftop solar plant installed by a residential consumer under simplified procedure

Plant Capacity	Applicable Subsidy
Up to 3 kW	Rs. 14588/- per kW
Above 3 kW and up to 10 kW	Rs. 14588/- per kW for first 3 kW and thereafter Rs. 7294/- per kW
Above 10 kW	Rs. 94822/- fixed

Component B(Incentives to DISCOMs): DISCOMs will receive incentives based on their success in facilitating rooftop solar installations up to a total of 18 GW capacity.

2. Pradhan Mantri Surya Ghar: Mufti Bijli Yojana [39][40]:

The objective of the scheme is to increase the share of solar rooftop capacity and empower residential households to generate their own electricity.

The scheme provides for a subsidy of 60% of the solar unit cost for systems up to 2kW capacity and 40 percent of additional system cost for systems between 2 to 3kW capacity. The subsidy has been capped at 3kW capacity. At current benchmark prices, this will mean Rs 30,000 subsidy for 1kW system, Rs 60,000 for 2kW systems and Rs 78,000 for 3kW systems or higher.

Table 2: Subsidies given to residential households

Subsidy for residential households

Rs. 30,000/- per kW up to 2 kW

Rs. 18,000/- per kW for additional capacity up to 3 kW

Total Subsidy for systems larger than 3 kW **capped at Rs 78,000**

Suitable Rooftop Solar Plant Capacity for households

Average Monthly Electricity Consumption (units)	Suitable Rooftop Solar Plant Capacity
0-150	1 – 2 kW
150-300	2 – 3 kW
>300	Above 3 kW

The majority (about 88%) of the scheme outlay of Rs 75,021 crore (US$8.97 billion) is assigned as direct CFA to residential consumers. About 8% of the outlay is designated to incentivise local bodies such as urban local bodies, gram panchayats and state DISCOMs to support the adoption of rooftop solar by residential consumers. The rest of the outlay is divided between enhancing awareness, capacity building, developing model solar villages, developing innovative projects and a payment security mechanism (PSM).

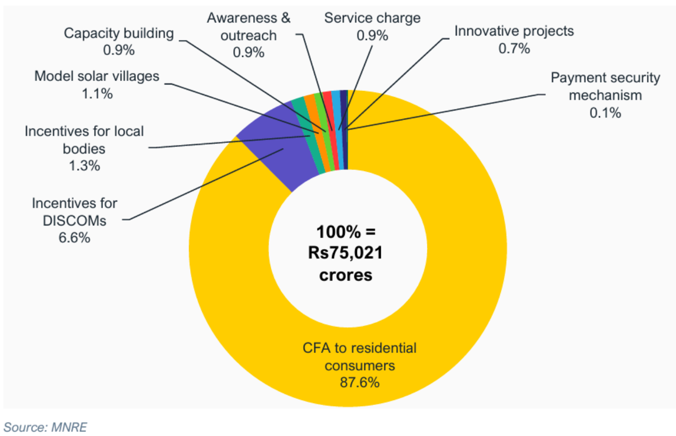

Figure 1: Graph showing PMSGY component wise financial outlay

3.2: Policies and Regulations in the assigned state of Kerala:

Kerala Solar Power Policy, 2013 [41]:

Kerala state sets the target to achieve 2500 MW of solar installed capacity by 2030 to contribute to the long term energy security of the Kerala as well as to the environment by reduction in carbon footprint. It defines user that can adapt solar and target them by deploying package of incentives which are discussed in the later part of this report. Kerala solar power policy uses a number of strategies for its successful implementation. Few important one are discussed in brief below.

1. Supply-Side Interventions:

The policy emphasizes promoting solar energy adoption through various supply-side measures. Off-grid rooftop systems are encouraged at consumer premises, including installations for solar inverters and cellular towers. *Grid-connected systems are promoted* to allow surplus energy to be fed into the grid. Off-site generation is encouraged in areas such as canals, reservoirs, waste lands, and quarries, while offshore solar-thermal generating plants are also considered. Diesel generators are to be replaced with off-grid solar applications, following MNRE guidelines. Additionally, suppliers and integrators for solar system installations will be empaneled, and standards for grid connectivity at LT levels will be developed until national standards are adopted.

2. Promotion of Solar Thermal Collectors:

The policy mandates the adoption of solar thermal technologies across various sectors. Solar Water Heating Systems (SWHS) are made mandatory for industries, hospitals, hotels, residential buildings exceeding 3000 sq. ft., hostels, and sports complexes. Solar steam systems are encouraged for community cooking in institutions like hospitals and mid-day meal programs, as well as for industrial applications such as textile and food processing industries. Furthermore, industrial applications requiring hot water, steam pre-heating, drying processes, and laundry units are encouraged to adopt solar technologies.

3. Financing Mechanisms:

Innovative financing mechanisms are proposed to support solar energy projects. Bank financing will be offered at attractive rates for off-grid systems instead of providing capital subsidies. Non-government buildings will receive incentives through net metering and feed-in tariffs. Cluster-wise installations will be incentivized under specific conditions, and community ownership models will be promoted for photovoltaic installations on water bodies.

4. Feed-in Tariff & Net Metering:

The Kerala State Electricity Regulatory Commission (KSERC) will play a key role in facilitating feed-in tariffs and net metering systems. Feed-in tariffs will be notified for offsite commercial installations, while net metering will be promoted for subsidized installations that consume grid power.

5. Infrastructure Development:

To ensure successful implementation, the policy focuses on strengthening infrastructure development. International safety and quality protocols will be established with the help of academic expertise. Grid quality will be improved through nano/community grids based on smart grid principles. Additionally, the policy integrates campaigns like "no load shedding" with solar energy initiatives to ensure reliable power supply.

Policy instruments of Kerala solar power policy [42]:

The state government deployed the SOURA program to encourage GRPV system in 2019. The program, a initiative to add 1000 MW of solar power to the grid, was launched in 2019-20. Phase 1 of the program was launched in 2019-20, while Phase 2 followed in November 2020.

Phase 1 targeted 200 MW of installation but only 26 MW was achieved and so it was modified to Phase 2 which was highly successful achieving a installation of 127.5 MW. This success is because of the Kerala Model which was appreciated by MNRE and World Bank as one of the best business model. This has been discussed in detail in the report.

SOURA Scheme [43]:

The bottleneck in solar systems is its high capital requirement which impacts particularly more to domestic consumers. To tackle this, Kerala State Electricity Board (KSEB) came up with this scheme of SOURA providing subsidies of up to 40% to the individual willing to install GRPV system on their rooftop. The KSEB created a portal named Ekiran where it streamlined the process of applications, mentioned requirements and conditions. SOURA provided subsidies as shown below but still it was not able to mitigate the issue of high initial capital.

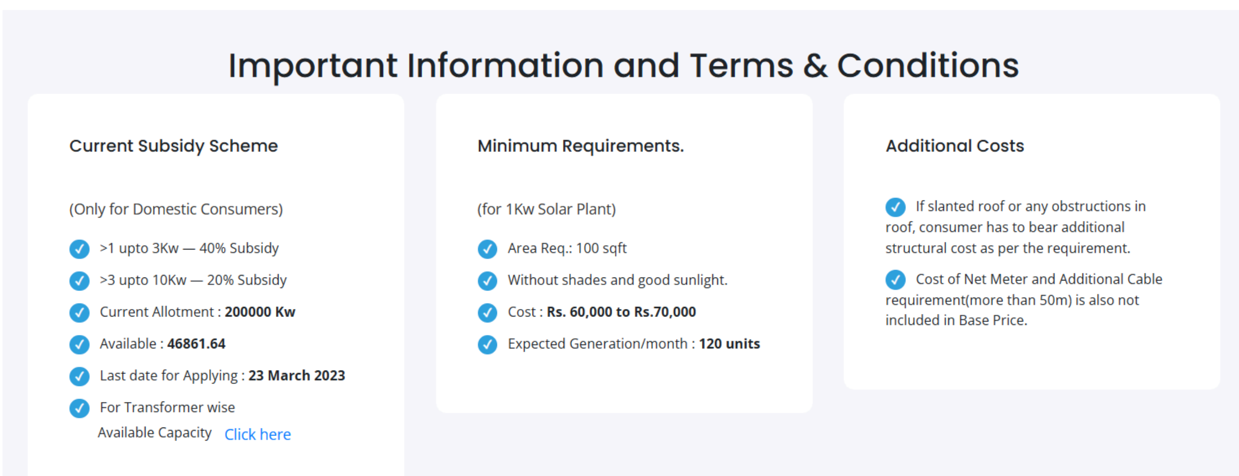

Figure 2: Terms and conditions for SOURA scheme

https://kseb.in/uploads/Downloadtemsuppy/VOL%20I%20Issue%205-16990941592043930070-17091222055391471\03.pdf

Kerala Model [45]:

In the phase 2 of SOURA program, the Kerala model came up with a very interesting models which made the Kerala to stand at 4th position among the Indian states having highest rooftop capacity as of 2024. Launched by the KSEB in Nov 2021, this program incentivizes rooftop solar adoption by covering installation costs. They categorized consumers in different brackets (1A, 1B, 1C) and provided financial assistance which was huge compared to the one of SOURA phase 1.

Model-1: In the model 1, consumer has to pay some percentage of the capital and in return they get double percentage of electricity produced by the system as shown in the table. Consumers rent their rooftops to KSEB, receiving some part of the generated solar power for free while KSEB manages operations and maintenance for 25 years and owns the system. This initiative has significantly reduced consumer risk and increased trust in solar energy.

Table 3: Kerala model [45]

Models	Average monthly consumption (units)	Consumer contribution (% of cost)	Return (% of plant generation)
1A	120	12%	25%
1B	150	20%	40%
1C	200	25%	50%

Model-2: KSEB will install a rooftop plant at consumer premises and the energy generated will be sold to the consumer at fixed price for 25 years through Purchase Price Allocation (PPA). KSEB will install and maintain the plant for 25 years, incurring the entire cost of installation.

Model-3: KSEB will set up the solar plant after collecting the cost of the plant from the consumer. Excess energy, if any, after the consumption of the consumer (plant owner) will be settled at APPC (Average Power Purchase Cost) rate approved by Karnataka State Electricity Regulatory Commission (KSERC) at the end of settlement period [46].

SOURA Tejas Program:

After the success of Kerala Model, KSEB decided to promote solar adoption among commercial, and institutional consumers as well while contributing to Kerala’s renewable energy goals. One of the key policies of the program is the implementation of the net metering mechanism, which allows consumers to feed excess solar energy into the grid and earn credits that can offset their electricity bills. Financial incentives, including subsidies of up to 40% for rooftop systems of up to 3 kW, are provided as per the guidelines of the Ministry of New and Renewable Energy (MNRE). Consumers can either opt for the self-owned model, where they install and own the solar system, or the KSEB-owned model, where KSEB installs the system and pays a rental fee for the rooftop space. The program ensures strict quality standards by engaging only KSEB-empaneled vendors for installation and maintenance. It sets capacity limits for residential consumers, typically ranging from 1 kW to 10 kW, while larger installations for institutional and commercial establishments require specific approvals. The policy of this program also focus on grid stability by limiting the cumulative solar capacity to 30% of the transformer's load capacity in any area.

A new statutory body comes into picture in SOURA Tejas program. The Agency for New and Renewable Energy Research and Technology (ANERT) plays a critical role in the implementation and promotion of the Soura Tejas Program in Kerala. As the nodal agency for renewable energy initiatives in the state, ANERT is responsible for policy formulation, awareness generation, and facilitating solar energy adoption among citizens. In the context of the Soura Tejas Program, ANERT works closely with the KSEB to ensure smooth execution, monitor quality standards, and coordinate the installation of rooftop solar systems [47].

Solar City Project:

The Solar City Project aims to transform the capital city Thiruvananthapuram into India's biggest solar powered city. A special attention has been given to grid-connected rooftop solar system. ANERT has identified more than 3 lakh buildings in Kerala and created a solar atlas which are suitable for installation of solar panel on roofs.

Under the Solar City Project in Kerala, consumers are offered a range of financial and non-financial incentives to encourage the adoption of solar energy. One of the key benefits includes up to 40% subsidy on the installation cost of rooftop solar systems, as per the guidelines of the MNRE under PM Surya Ghar Yojana. ANERT also provides a discount of 4% on the interest of bank loans for residential rooftop solar systems.The net metering mechanism is another incentive, allowing consumers to earn credits for surplus electricity fed back into the grid, which can be adjusted against future electricity bills. Consumers in areas with limited grid connectivity are provided financial support for off-grid solar systems, and low-interest loans are also facilitated to reduce the burden of upfront installation costs. Moreover, incentives are extended to public institutions, including government offices, schools, and hospitals, to encourage solar adoption for significant energy savings.

3.3. Policies and Regulations in Telangana [48]

The Telangana State Electricity Regulatory Commission (TSERC) introduced the Net Metering Regulation in 2016 (Regulation No. 6 of 2016) to promote the adoption of grid-connected rooftop solar photovoltaic (PV) systems across the state. This policy enables consumers to generate their own solar electricity and offset their consumption by feeding surplus power back into the grid, thereby encouraging decentralized renewable energy generation, reducing dependence on fossil fuels, and supporting sustainability goals.

Eligibility and Scope:

- Applicable to all electricity consumers of distribution licensees in Telangana.
- Allows rooftop solar PV systems under both ownership and third-party models.

System Capacity:

- Minimum capacity: 1 kWp; Maximum capacity: 1 MWp.
- Residential/Government consumers: Up to 100% of sanctioned load.
- Commercial/Industrial consumers: Up to 80% of sanctioned load.

Metering Requirements:

- Mandatory use of bi-directional (net) meters as per CEA (Central Electricity Authority) standards.
- Solar generation meter also required for Renewable Purchase Obligation (RPO) accounting.

Energy Accounting and Settlement:

- Net energy is calculated by adjusting exported units against imported units monthly.
- Surplus energy is carried forward for six months and settled at the Average Pooled Power Purchase Cost (APPC) at the end of each settlement period.

Connection and Safety Standards:

- Compliance with CEA connectivity and safety guidelines is mandatory.
- Includes anti-islanding protection, voltage/frequency trip settings, and other grid safety mechanisms.

Exemptions and Incentives:

- Exempted from wheeling charges, transmission losses, and cross-subsidy surcharges.
- Eligible for central and state subsidies/incentives.
- Provision to avail Clean Development Mechanism (CDM) benefits.

System Validity and Agreement:

- Net metering agreement remains valid for 25 years or system life, whichever is earlier.
- Net and gross metering cannot be availed simultaneously at the same premises

Key Additions in the 2020 Guidelines (TSSPDCL) over the 2016 Regulation [49]:

Simplified Application Process

- Online and offline submission options introduced.
- Clear list of documents and fixed application fees.

Faster Approvals

- Feasibility approval timeline reduced from 21 to 7 working days.
- Deemed approval if delayed.

Stricter Grid Capacity Limits

- Cumulative solar capacity limited to 50% of transformer capacity.
- Clear MW caps defined for 11 kV and 33 kV feeders.

Detailed Metering Requirements

- Mandatory Advanced Metering Infrastructure (AMI) for systems >10 kW.
- Specified accuracy classes for different system sizes.

3.4. The metering and billing mechanisms and their impact on promoting these systems in Kerala [50]:

3.4.1. Overview of Net Metering

Net metering is a billing mechanism that allows residential consumers who generate electricity using rooftop solar photovoltaic (PV) systems to export surplus energy to the grid. This mechanism enables the consumer (prosumer) to consume self-generated electricity first, and send any excess electricity to the grid. In periods where the prosumer's generation is insufficient, electricity is imported from the grid. Net metering promotes self-reliance, reduces electricity bills, and supports the use of clean energy.

3.4.2. Net Meter Installation and Function

A bi-directional net meter is installed at the interconnection point between the prosumer’s system and the distribution network. This meter records both the import of energy from the grid and the export of energy to the grid. The metering arrangement must comply with the standards laid down by the Central Electricity Authority (CEA). In the event that the distribution licensee cannot supply a net meter within 10 days of request, the consumer may purchase a meter independently, ensuring it meets the required specifications.

3.4.3. Billing Period and Net Energy Calculation

Electricity bills are issued for each billing cycle as determined by the distribution licensee. During each billing period, the net energy is calculated by deducting the total exported energy from the total imported energy. If the prosumer has consumed more electricity than exported, they are billed for the net import. If the exported energy exceeds consumption, the excess is banked and carried forward to the subsequent billing cycle.

3.4.4. Time-of-Day Adjustments and Settlement for Consumers Above 20 kW

For prosumers with connected loads above 20 kW, energy accounting includes time-of-day (ToD) considerations, dividing the day into peak, normal, and off-peak hours. Energy exported during one time block can be adjusted against energy imported in another, with specific adjustment rates. For instance, 80% of energy exported during normal hours can be adjusted against peak hour consumption. Likewise, energy exported during peak hours may be adjusted 100% during peak and 120% during other hours.

3.4.5. Yearly Settlement and Financial Compensation

At the end of the financial settlement period, if there is any net surplus energy remaining in the consumer's account, the distribution licensee is required to compensate the prosumer at the Average Power Purchase Cost (APPC) as approved by the Kerala State Electricity Regulatory Commission (KSERC). If payment is delayed beyond 30 days from the settlement date, interest shall be paid at the rate of FBIL + 200 basis points prevailing on the 1st of April of the settlement year.

3.4.6. Exemptions and Incentives

Prosumers under net metering are exempted from transmission charges, wheeling charges, and cross-subsidy surcharges for electricity generated and consumed within the same premises. Additionally, the quantum of electricity generated through the rooftop solar system and consumed at the premises qualifies towards fulfilling the Renewable Purchase Obligation (RPO) of the distribution licensee.

3.4.7. Impact of Net metering

The implementation of net metering in Kerala has had a significant positive impact on the state's energy landscape. It has encouraged thousands of households to install rooftop solar systems, contributing to increased adoption of renewable energy. This has helped in reducing dependence on conventional fossil fuel-based electricity and supported the state's goals for sustainability and energy security. Economically, it has empowered consumers by reducing their electricity bills and enabling them to become energy producers. Environmentally, the increased use of solar energy has led to lower carbon emissions. Moreover, the electricity grid has benefited from decentralised power generation, which helps in reducing transmission losses and improving grid reliability. Overall, net metering has played a crucial role in promoting clean energy in Kerala while engaging citizens directly in the energy transition.

4. Stakeholder Engagement

4.1 Individual Stakeholder Insights other than consumers

4.1.1 Solar Technician (Interviewed by Pooja as per Appendix A) :

The technician is a person based in a rural area who repairs solar GRPV systems. Based on the questionnaire prepared, the following insights have been gained from the interview.

Need for User Awareness: Emphasized the importance of basic knowledge among users regarding the regular maintenance of solar systems to ensure longevity and optimal performance.
Panel Maintenance: Highlighted the necessity of frequent panel cleaning to maintain efficiency.
Battery and Inverter Risks: Noted that sudden drops in power output can lead to damage or burnout of system components like the battery and inverter.
Grid Connectivity: Recommended integrating grid connections to reduce dependency on batteries, especially during low-sunlight periods.
Technician Accessibility: Stressed the lack of trained technicians in rural areas, often resulting in delays and the need for long-distance travel to urban centers for repairs.
Timely Repairs: Advised prompt attention to minor issues to prevent them from escalating into major problems, which are costlier and harder to fix.
Preventive Approach: Urged users not to wait until system failure; early intervention is both more manageable and economically efficient.

4.1.2 Solar Installer (Interviewed by Sanket Mhaske as per Appendix B) :

Based on the questionnaire prepared, following valuable insights are shared by the solar installer who has experience of 4 years in rooftop solar installation. The solar installer holds a diploma in electrical engineering and undergo training happening throughout the year related to solar technology.

4.1.2.1 System Sizing Based on Consumption

Solar installers generally follow a rule-of-thumb approach to determine the appropriate system capacity based on a household's monthly electricity consumption:

- 0–100 kWh → 1 kW system
- 100–200 kWh → 2 kW system
- 200–300 kWh → 3 kW system

4.1.2.2. Cost Sensitivity and Panel Quality

The overall cost of a rooftop solar system is highly influenced by the brand and quality of solar panels selected. While higher quality panels may increase the initial capital expenditure (CAPEX), they typically come with better warranties and after-sales service. Installers advise customers to focus on long-term benefits rather than opting for cheaper alternatives that may compromise performance and durability.

4.1.2.3. Physical Challenges and Installation Barriers

Installers report that space availability and shading are among the most common challenges faced during rooftop solar installations. Shading often results from nearby trees, which may lead to tree cutting. Additionally, the presence of water tanks or other rooftop structures can create installation hurdles, necessitating a careful selection of alternate locations for panel mounting.

4.1.2.4. Mounting Structures and Elevation Considerations

In most cases, panels are installed at a standard elevation to optimize solar generation. However, when rooftops already have built-in structures or when customers seek to avoid additional costs for mounting frameworks, panels are installed directly on existing structures. This can potentially affect the efficiency of power generation due to suboptimal tilt or orientation.

4.1.2.5. Role of Subsidies and User Perception

While government subsidies contribute to encouraging adoption, the primary motivator for most households remains the elimination of electricity bills after installation. As expressed by the installer, "Customers will not be happy even if you give everything free to them," unless they see tangible financial benefits. Long-term warranties of 25–30 years also provide assurance, especially to male members of the household, while housewives often express satisfaction due to the monthly cost savings.

4.1.3. MSEDCL Assistant engineer (Interviewed by Aditya Vibhute as per Appendix C) :

The assistant engineer is working for MSEDCL Baramati and Indapur divisions. He has been working for MSEDCL for more than 14 years. His work mainly consists of maintaining power cables, checking for AT&C losses and going to places if there is a power disruption and analyzing the issue.

4.1.3.1. GRPV system Connection to the Grid:

A household with a rooftop solar PV system is connected to the DISCOM via two meters:

1. A net meter: Measures electricity imported from and exported to the grid.
2. A generation meter: Measures electricity generated by the rooftop solar system.

The electricity generated is converted through an inverter and fed into the nearest distribution pole, which connects it to the grid.

4.1.3.2. Net Billing Mechanism:

1. At the end of every financial year (1st April), the difference between electricity generated and imported is calculated.
2. If generation exceeds import, the excess is compensated at approximately Rs. 2 per kWh, which is credited in the consumer's electricity bill for the next month.
3. Over time, this can result in zero-amount electricity bills for prosumers if the generation consistently exceeds consumption.

4.1.3.3. Monitoring of Electricity Consumption:

Consumption is monitored by the transmission department using systems such as SCADA (Supervisory Control and Data Acquisition), which displays various operational parameters on a centralized screen.

4.1.3.4. Perspective on Rooftop Solar Installations:

The engineer recommends rooftop solar installations, emphasizing long-term savings.
He suggests people should consider installing them if they have sufficient space and financial means.

4.1.4. Solar Installer (Interviewed by Sanket Bafana as per Appendix D)

The interview with Mr. Gajendra Sulakhe, Manager at PRE RAYS ENERGIES, highlights strong growth in India's GRPV sector, driven by rising electricity costs, improved financing, and supportive government schemes like the Rooftop Solar Programme Phase II. With over 500 installations across residential, commercial, and industrial segments, the company has seen increasing adoption due to benefits like reduced electricity bills, shorter payback periods, and energy independence. However, challenges persist in the form of high upfront costs, limited awareness of financing options, and policy inconsistency across states. While residential users benefit from subsidies, commercial and industrial segments lack direct incentives, relying more on financing models like Power Purchase Agreements.

Technical hurdles such as roof space, structural limitations, and shading are common, though advancements in panel efficiency and smart inverters are helping. The sector is expected to grow at 20–25% annually, but key issues like delays in net metering approvals, DISCOM resistance, and uneven state policies need to be addressed. Overall, a stable regulatory environment and wider access to financing are essential to accelerate GRPV adoption and ensure long-term growth.

4.2 Insights from Consumer Survey:

4.2.1. Consumer Survey Insights (Surveyed by Sanket Mhaske):

This section outlines insights derived from interviews with three consumers who have installed on-grid solar photovoltaic (PV) systems without battery storage. All three respondents are engineers employed at HAL and reside in individual bungalows. The study focuses on their motivations, experiences post-installation, concerns, and reflections.

Consumer 1: Engineer, Ozar

Insights:

- The consumer was motivated to install the system after learning about the savings experienced by a friend.
- He had prior knowledge of the economic viability of the system despite the high initial capital expenditure (CAPEX).
- Post-installation, the entire electrical load is met by the solar system.
- The respondent expressed concern over the low feed-in-tariff (Rs. 3 to Rs. 3.5/unit), especially compared to the grid electricity rate of Rs. 11–12/unit.
- He seeks a more equitable compensation structure for surplus power supplied to the grid.

Reflection:

- A sound understanding of the financial aspects, especially the payback period, helped the respondent make a confident investment decision.
- His technical background enabled him to appreciate long-term benefits over short-term costs.

Consumer 2: Engineer, Nashik

Insights:

- Inspired by a peer's experience, the consumer installed a solar system to reduce electricity expenses.
- The system fulfills his entire monthly electricity demand.
- He is concerned by recent reports suggesting that MSEB may begin charging consumers for electricity used between 6 p.m. and 9 a.m.
- The consumer wished to install a higher-capacity system to accommodate future load but was restricted by rooftop space due to existing installations (solar water heater and water tank).

Reflection:

- The respondent’s anxiety highlights the need for stable and consumer-friendly solar energy policies to encourage adoption.
- Despite constraints, his decision demonstrates forward-thinking and a commitment to sustainable energy.

Consumer 3: Engineer, Nashik

Insights:

- The consumer demonstrated a proactive approach to system maintenance by installing taps connected to each module for ease of cleaning.

Reflection:

- His attention to detail reflects a high level of engagement with the technology, which is crucial for maintaining long-term system efficiency.

Common Insights Across All Respondents:

- All consumers have installed on-grid systems without battery storage, rated at 3 kW.
- Prior to installation, monthly electricity usage exceeded 200 units.
- Each respondent reported a reduction of 80–90% in their electricity bills post-installation.
- Regular maintenance is practiced, with panel cleaning conducted every 15–20 days using water.
- Total project costs ranged from Rs. 190,000 to Rs. 215,000, with approximately Rs. 78,000 borne by the consumers after subsidy.

- All were influenced by peer experiences and saw the system as a financially viable investment.
- The first and third consumers increased the use of electric appliances (e.g., electric water heaters, induction cookstoves), while the second already utilized solar water heating and added an induction stove post-installation.

Common Reflections:

- All respondents share similar socio-economic backgrounds: engineers employed at HAL, residing in bungalows, and possessing the financial capacity to self-fund their installations.
- Peer influence within their professional circle played a pivotal role in decision-making.
- While government subsidies were beneficial, the primary driver for adopting solar PV systems was the high cost of conventional electricity and the associated potential for long-term savings.

4.2.2. Consumer Survey Insights (Surveyed by Sanket Bafana):

This section highlights insights from two urban consumers in Dhule who have installed on-grid solar PV systems for residential and commercial purposes. Their experiences provide valuable information on technical understanding, financial outcomes, and procedural challenges.

Consumer 1: Urban Residential + Commercial (Shop), Dhule

Reason for Installation:

- Primary motivation was to reduce high electricity expenses for both his residence and shop.

System Details:

- Capacity: 6.3 kW
- Usage Duration: 3 years
- Subsidy Received: Rs. 1,08,000 (comprising Rs. 78,000 + Rs. 30,000)
- Average Monthly Generation: 560 units
- Average Monthly Export to Grid: 240 units
- Monthly Bill Reduced: From Rs. 6000 to Rs. 0

Experience:

- Installation decision was influenced by recommendations from friends and direct interaction with the installer.
- Achieved zero billing just two months after installation; surplus energy exported to the grid.
- Faced delays in subsidy disbursement and extensive paperwork; however, these were managed by the installer.

Technical Observations:

- Demonstrates a strong understanding of the net metering and system operations.
- The installed system is slightly oversized relative to consumption, leading to excess generation and potential overinvestment.

Follow-up:

- Respondent provided the February 2025 electricity bill upon request. Earlier bills, including pre-installation data, were not available.

Reflection:

- The respondent is highly satisfied with the outcome despite initial paperwork hurdles.
- Highlights the importance of accurate system sizing based on energy needs.

Consumer 2: Urban Residential + Commercial (Hospital), Dhule

Reason for Installation:

- Installed to offset substantial electricity bills incurred by both residence and hospital operations.

System Details:

- Capacity: 25 kW
- Usage Duration: 2 years
- Subsidy: Not availed (Consumer was unaware of the option)
- Monthly Bill: Reduced to zero

- Maintenance: Experienced a single inverter malfunction which was repaired without replacement.

Experience:

- Chose installation based on peer recommendation and installer guidance.
- Achieved zero billing within two months; exports surplus energy to the grid.
- Opted not to apply for a subsidy due to lack of awareness at the time of installation.

Technical Observations:

- Fully understands net metering and system functionality.
- The consumer’s decision to proceed without subsidy shows strong financial confidence but also points to a lack of information dissemination regarding government schemes.

Follow-up:

- Shared the electricity bill for February 2025. Pre-installation bill records were not available.

Reflection:

- Satisfaction with performance and return on investment is evident.
- Highlights a need for better outreach and awareness regarding available financial incentives like subsidies.

Key Takeaways from Urban Dhule Respondents:

- Both respondents achieved 100% savings on electricity bills within two months of installation.
- Peer influence and installer credibility were crucial in decision-making.
- Both users have a strong technical understanding of their systems and the net metering process.
- Challenges included delays in subsidy processing and lack of awareness regarding subsidy options.
- One consumer's system appeared oversized, indicating the importance of precise load assessment during system planning.

4.2.3. Consumer Survey Insights (Surveyed by Aditya Vibhute):

This section highlights insights from three consumers located in rural and semi-urban areas who have installed on-grid solar photovoltaic (PV) systems. The respondents' experiences provide valuable perspectives on the motivations, challenges, technical knowledge, and financial outcomes related to their solar systems.

Consumer 1: Rural, Basti, Uttar Pradesh

Reason for Installation:

- The consumer sought to reduce electricity bills, contribute positively to the environment, and lower carbon dioxide emissions.

System Details:

- Usage Duration: 1 year
- Subsidy: Rs. 1,08,000 (Rs. 78,000 + Rs. 30,000)
- Monthly Savings: 80% reduction from Rs. 2000 to Rs. 400
- Experience: System resets when power generation is very high, and this process can take up to one hour.
- Aesthetic improvements were suggested to enhance the system's appearance.

Experience:

- The consumer was referred by friends, family, and neighbors and had direct discussions with the installer.
- The system generated high savings and the consumer appreciated the net metering mechanism.
- Faced challenges with the subsidy process, which involved extensive paperwork and delayed approvals.

Insights from Experience:

- The consumer emphasized the importance of installing an on-grid system due to the absence of subsidies for off-grid systems.
- The high cost of battery storage was noted as a deterrent.

Technical Observations:

1. The respondent fully understood the net metering system and bill payment process.
2. The system's function was explained in simple terms to the consumer.
3. The consumer prioritized installing a grid-connected system, which they considered more efficient than off-grid options.
4. Cost-saving was the primary motivator, with environmental concerns as secondary.

Social Observations:

- The respondent, being literate, had a solid grasp of the system's technical aspects and was able to respond to queries with ease.

Follow-up:

- The consumer shared only the October 2024 electricity bill. The request for additional photos was met with reluctance, possibly due to unfamiliarity with the requester.

Reflection:

- The respondent appears satisfied with the savings but showed frustration with the documentation process and the additional photo requests.

Consumer 2: Semi-Urban, Pandharpur, Maharashtra

Reason for Installation:

- Motivated by the desire to use clean energy, reduce pollution, and rely on an independent power source instead of coal-based electricity.

System Details:

- Usage Duration: 5 years
- Subsidy: Rs. 68,000
- Monthly Savings: 97% reduction, from Rs. 3500 to Rs. 100
- Maintenance: Cleaning once a month during summer, no cleaning in the monsoon, and once every two months in winter.

Experience:

- The consumer explored options through an online forum and found the installer in Solapur.
- The system has proven highly effective, delivering substantial savings on electricity bills.
- Faced delays in both approval and subsidy disbursement.

Insights from Experience:

1. The consumer recommended installing monocrystalline solar panels for better efficiency.
2. The consumer also installed solar panels for a 3 HP water pump for irrigation, which resulted in zero electricity bills.

Technical Observations:

1. The consumer was very pleased with the savings and found the system's performance to be highly efficient.
2. The consumer became so convinced of the benefits of solar energy that they installed solar panels on their farm as well.

Social Observations:

- The consumer, being literate, understood the technical workings of the system and provided comprehensive responses to questions.

Follow-up:

- The respondent was very cooperative, promptly providing all requested photos, including those of the meter, inverter, PV panels, and the electricity bill.

Reflection:

- The consumer was highly motivated by the savings and expressed strong enthusiasm for encouraging others to adopt solar energy.

Consumer 3: Rural, Varanasi, Uttar Pradesh

Reason for Installation:

- Installed the system for cheaper electricity, independence from the grid, and greater savings.

System Details:

- Usage Duration: 10 months
- Subsidy: Rs. 90,000
- Monthly Savings: 90% reduction in electricity bills.

Experience:

- The system was installed through a local installer, recommended by a family friend.
- Cleaning the panels is challenging due to restricted access with only a ladder available.

Insights from Experience:

1. The consumer was particularly motivated by the potential cost savings.
2. The consumer was unaware of specific details about the inverter but showed interest in learning more.
3. More efficient panels and quicker government response for subsidies were suggested for improvement.

Technical Observations:

1. The consumer was focused primarily on the financial benefits, with limited understanding of system specifics.
2. The family was involved in the decision-making, with the respondent actively proposing the idea.

Social Observations:

- The consumer demonstrated a basic understanding of the system but did not have detailed knowledge about the inverter or broader financial aspects.

Follow-up:

- The consumer sent photos of the PV panels and electricity bill, but did not provide photos of the meter or inverter plate.

Reflection:

- The consumer expressed satisfaction with the system's performance but highlighted cleaning difficulties and delays in receiving subsidies.

Key Takeaways from Rural and Semi-Urban Respondents:

- Motivation for installation primarily revolved around cost savings and environmental concerns, with the latter being secondary for most.
- All respondents were literate and had a reasonable grasp of the technical aspects of their systems.
- Subsidy delays and long approval times were consistent challenges.
- Two of the respondents also highlighted the aesthetic appeal of the systems and possible improvements in efficiency (e.g., bifacial panels, more efficient solar technology).
- One respondent also extended solar adoption to their farm operations, demonstrating the scalability of solar energy solutions.

4.2.4. Consumer Survey Insights (Surveyed by Pooja Mathary):

This section presents insights from three consumers using off-grid solar photovoltaic (PV) systems in rural and urban areas. The consumers' experiences highlight the challenges they faced with system reliability, maintenance, and the shift to grid-connected systems.

Consumer 1: Rural, Telangana

Reason for Installation:

- Installed solar system due to inconsistent power supply in the village.

System Details:

- Usage Duration: 4 years
- Experience: The inverter failed during the usage period, and no support was received from the installer. As a result, the consumer switched to the main grid.

Reasons for Stopping Use:

- Lack of post-installation repair service was the primary reason for discontinuation.

- The only repair option was to go to a city for servicing, which was not practical.
- Additionally, the government provided free electricity up to 200 units, removing any incentive to repair the system.

Future Plans:

- Would consider using solar only as a supplement to the main grid, ideally with a grid-connected system.
- Complaints about the quality of the system and a lack of reliable vendors made him hesitant to trust any future solar system installations.

Reflection:

- The consumer valued the use of solar when it was functional but highlighted the importance of after-sales service and the need for accessible repair options.
- The introduction of free electricity from the government influenced the decision to abandon solar for the time being.

Consumer 2: Rural, Telangana

Reason for Installation:

- Installed the system to address inconsistent power supply in the village.

System Details:

- Usage Duration: 8 years
- Experience: The inverter failed multiple times, and repairs took place in the city, usually taking around 7 days each time.
- Eventually switched to the main grid but had been happy using the solar system when it worked.

Reasons for Stopping Use:

- The key issue was the repair service post-installation, which required traveling to the city.
- Despite efforts to continue using solar, the introduction of free electricity up to 200 units by the government led to the decision to stop using the solar system.

Future Plans:

- The consumer is interested in using solar in the future, but would prefer a grid-connected system.
- He expressed frustration with the limited number of distributors, which restricted options for purchasing and maintaining solar systems.

Reflection:

- The consumer appreciated solar energy when it was functioning but felt that the lack of repair support, combined with government subsidies, reduced the motivation to continue using off-grid solar.
- Grid connection and the availability of more distributors were important considerations for any future solar installations.

Consumer 3: Urban, Maharashtra

Reason for Installation:

- Installed solar to reduce electricity bills.

System Details:

- Usage Duration: Ongoing
- Experience: The system is functioning well with minimal maintenance. Inverter repairs can be completed in 1-2 days, an advantage due to easy accessibility.
- Savings of 20-25% on electricity bills.

Current Status:

- The consumer is still using the solar system and plans to increase its capacity for further savings.

Reflection:

- The consumer is satisfied with the performance and maintenance of the system, which is less prone to major issues due to better accessibility.

- The savings on bills have been substantial, and the consumer is looking to expand the system for increased benefits.

Key Insights and Learnings:

- Maintenance and Repair Services: A consistent theme among the rural respondents was the lack of adequate post-installation support, especially regarding repairs. This issue influenced their decision to move to grid-connected systems.
- Government Subsidies: The introduction of free electricity up to 200 units by the government played a significant role in reducing the motivation to maintain off-grid solar systems.
- Grid-Connected Systems: Both rural respondents expressed a preference for grid-connected solar systems in the future, citing the need for reliable repair services and the benefits of combining solar with the main grid.
- Urban Experience: In contrast, the urban consumer in Maharashtra has had a positive experience with solar, benefiting from easy accessibility for repairs and significant savings on bills. This consumer is interested in expanding the system for further financial benefits.

4.3 Consolidated Findings from Stakeholder Interviews and Consumer surveys:

4.3.1. Financial Viability and Impact

- Consumers: Across urban, semi-urban, and rural settings, users consistently reported significant bill reductions (80%–100%), with some achieving zero billing within two months of installation.
- Stakeholders (Installers/DisComs): Affirmed that GRPV systems provide attractive payback periods, especially when subsidies are availed.
- Common View: Financial savings are the primary motivator for adoption. However, both groups flagged the complexity and delay in subsidy disbursement as a deterrent.

4.3.2. Role of Subsidies

- Positive Financial Support: Where availed, subsidies ranging from ₹68,000 to ₹1,08,000 significantly reduced upfront costs.

- Challenges in Processing: Many respondents experienced delays, extensive paperwork, and lack of clarity in subsidy disbursement.
- Lack of Awareness: Some consumers, especially in urban areas, missed out on subsidies due to lack of information, highlighting the need for better outreach and awareness.

4.3.3 Motivations for Installation

- Primary Driver – Financial Savings: Across the board, reducing electricity bills was the dominant motivator.
- Secondary Drivers:
  - Environmental Concerns were cited more often in semi-urban and rural areas.
  - Some also desired independence from coal-based electricity or unreliable local grids.
  - Peer recommendations and installer credibility played a critical role in decision-making.

4.3.4. Maintenance & Post-Installation Support

- Urban Advantage: Urban consumers had better access to repair services, ensuring quick turnaround for inverter or system issues (1–2 days).
- Rural Disadvantage: Rural users faced significant challenges with:
  - Inverter breakdowns.
  - Lack of local repair infrastructure, often requiring trips to the city.
  - Poor after-sales service, leading some to abandon solar systems.
- User-Initiated Maintenance: Basic cleaning was done by most users, especially in rural/semi-urban areas.

4.3.5. Net Metering Experience

- Positive Perception: Almost all GRPV users expressed satisfaction with the net metering process, noting it enabled surplus energy export and credit.
- Understanding Varies: Urban users had a stronger grasp, while some rural users only understood the billing outcomes, not the technicalities.

4.3.6. Social Influence & Literacy

- Word-of-Mouth & Peer Influence: Recommendations from friends, family, and neighbors were pivotal in system adoption.
- Literacy Correlation: Higher literacy levels generally led to better understanding of technical and financial aspects, and more proactive engagement.

4.3.7. System Scalability & Future Plans

- Some users, especially those with positive GRPV experiences, have already extended or plan to extend solar adoption to:
  - Farms (e.g., water pumps).
  - Additional residential or commercial loads.

4.3.8 Key recommendations based on Findings

1. Strengthen Consumer Education (especially in rural areas) about maintenance, performance expectations, and warranties.
2. Promote Financing Awareness: Leverage community outreach to introduce EMI schemes, PPAs, or leasing models.
3. Invest in Local Technician Training to improve service accessibility in underserved areas.
4. Enhance Peer-Led Campaigns by sharing successful case studies through social and professional networks.
5. Empower local installers and vendors through training and certification to enhance service quality and trust.
6. Mandate service quality metrics in vendor agreements for systems deployed under government subsidies.

5. Design of GRPV Systems and Their Financial Viability

Assumptions:

- HOMER is used to calculate LCOEs; all systems are optimal as suggested by HOMER.
- System capacity and inverter rating are calculated using the method mentioned in slides.
- This is a grid-connected system, so battery storage is not considered.

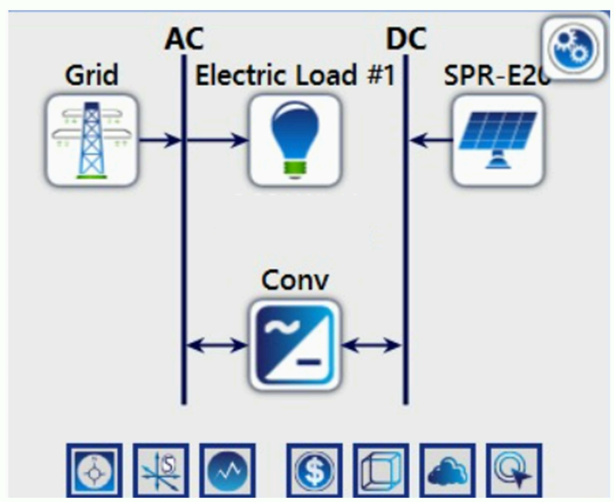

Figure 3: HOMER system configuration

Economic Assumptions:

- IRR: 6%, Inflation Rate: 5.53%
- Solar Panel Cost: ₹18,000 (500W panel)
- Inverter Cost: ₹16,000/kW, Installation: ₹10,000/kW

GRPV Design Summary:
https://docs.google.com/document/d/1sOMOdn_amDg1kLp8YXd3mNbW1sImVgk3/edit?usp=drive_link&ouid=105167289395469801118&rtpof=true&sd=true

Table 4: Technical and economical details from HOMER

User	Monthly Consumptions (kWh)	System Capacity (kW)	Inverter Rating (kVA)	Panels (500W)	Total Cost (₹)	Subsidy (₹)	Payback Period (Years)	
Aditya	245	2.5	3.5	5	1,71,000	90,000	4.83	4.79 / 2.4
Pooja	80	1	1.6	2	62,000	30,000	5.36	8.6 / 5.16
Sanket Bafana	50	0.5	1.5	1	47,000	15,000	2.1	11.3 / 7.5
Sanket Mhaske	270	2.5	3.6	5	1,72,600	69,000	5.95	6.38 / 4.27

Calculations:

Sanket Bafana: https://docs.google.com/document/d/1wWEtKpDFyop06Vx9Wa3K1YKZ8MqkBAxD/edit?usp=sharing&ouid=105167289395469801118&rtpof=true&sd=true

Aditya Vibhute: https://docs.google.com/document/d/1uC6G_yVcaw90-3tiVBL_ypeLxTjy-PBB/edit?usp=drive_link&ouid=105167289395469801118&rtpof=true&sd=true

Pooja M: https://docs.google.com/document/d/1lV0k1CZQusIvwiRJ004-ERx_LQl3AKhO/edit?usp=sharing&ouid=105167289395469801118&rtpof=true&sd=true

Sanket Mhaske: https://docs.google.com/document/d/1NWi-0sGFenK73tDbbCzu0IgIc56kbCkg/edit?usp=drive_link&ouid=105167289395469801118&rtpof=true&sd=true

6: Conclusion and Policy Recommendations

The two main challenges with rooftop solar systems are high initial capital and maintenance of the system. The first issue hinders the deployment of rooftop solar among the low-income consumers and the second issue of maintenance creates problems among the middle-class householders who are required to take care of the system. Even if the government provides subsidies and the consumers get solar systems installed on their rooftop, if not taken care by cleaning and other maintenance the efficiency of the system can significantly go down or they may defunct.

GRPV systems account for 72% of Kerala's total solar capacity, significantly higher than global averages. Kerala has proven that policy instruments when deployed by addressing the root cause of the problem can benefit the stakeholders significantly. These 2 major issues were addressed by Kerala using the Kerala Model which was built in the Phase 2 of SOURA program. Kerala has proposed DISCOM-centric business models, wherein the residential rooftop solar systems are either entirely or partially owned by the local DISCOM. Other states can also learn and improve their solar rooftop capacity using this model.

Policy of DISCOM-centric models:

The state electricity distribution company (DISCOM) is more closely associated with the consumer than any other organisation or institution promoting rooftop solar. It can be the best player to make the state's solar power capacity grow. It should propose different models by categorizing the consumers in classes such as low-income, middle-class, institutional, commercial, etc. rather than implementing only one type of model like PM Surya Ghar Yojana which seems viable mainly for middle-class households having high electricity bills.

Lower-income households can benefit from giving subsidies up to 60% or by giving incentive as in Kerala Model 1. Middle-class households have the problem of high electricity bills but are afraid to trust solar solutions. Their trust on solar can be built up if the DISCOM comes front and assures the maintenance of the system. For institutional and commercial ventures, it can provide financial assistance by tying up with the banks to provide discounts on interest rates and smoothly enable the loan facility.

The DISCOM can act as a demand aggregator which means it identifies the consumers from different categories having potential to adopt rooftop solar. This will also help the government by combining the demand of multiple buyers into one large order, it enables bulk purchasing, which significantly reduces the per-unit cost of equipment and installation. This collective approach increases bargaining power, allowing DISCOM to negotiate better deals on warranties, maintenance, and service quality. It also helps standardize the technical specifications and quality of installations across all participating households or buildings. With a single tender covering many users, vendor selection becomes more streamlined and competitive, leading to improved services. Additionally, financial institutions and government agencies find it easier to process loans and subsidies for aggregated projects, reducing administrative hurdles. The approach also allows for faster implementation and encourages community participation, building trust and awareness among potential adopters.

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Appendice

Appendix A:

Questionnaire for interviewing solar technician:

1. What are the most common types of repairs in GRPV systems?
2. What measures should be taken by the consumer to prevent frequent repairs?
3. What is the importance of maintenance and cleaning and how frequent should it be?
4. What is the main concern of people in rural areas to not use the system?
5. What measures should be taken by the government to encourage its usage?

Appendix B:

Questionnaire for Interviewing Solar Installer :

1. How many years of experience do you have in this field?
2. What is your educational background?
3. Have you undergone specific training for solar installations?
4. Do you have tie ups with any solar panel manufacturing company?
5. What's the smallest and largest residential solar system you've ever installed on a rooftop?
6. How do you determine what size solar system a customer needs?
7. What factors affect the cost of a solar installation the most?
8. Follow up to 7 : Why should customers go for the costlier/branded solar panels available?
9. What are the challenges or obstacles you face during installations?
10. How do you decide the tilt angle of solar panels?
11. How important are government subsidies in a customer's decision to go solar?

Appendix C:

Questionnaire for interviewing MSEDCL engineer:

1. What is your role and job at MSEDCL and how long have you been working there?
2. What is the difference between MSEB and MSEDCL?
3. How is connection made between a house which has rooftop solar PV installed and DISCOM?
4. How does the Net billing mechanism work?
5. How do you monitor the consumption of electricity?
6. How is the power generation from the solar power plant managed?
7. Do you think that people should install rooftop solar panels?

Appendix D:

Questionnaire for interviewing Solar installer Company manager:

1. Can you tell us about your company's experience in the GRPV sector?
2. What are the main types of GRPV systems that your company installs (residential, commercial, industrial)?
3. How has the demand for GRPV systems evolved in the past five years in India?
4. What are the key benefits of GRPV systems for customers?
5. Are there any specific government incentives or subsidies that have helped the GRPV sector grow?
6. What challenges do you face in terms of compliance and approvals from government authorities?
7. What are the common technical challenges in installing and maintaining GRPV systems?
8. How does net metering work in Maharashtra, and what challenges do you face in its implementation?

Additional results/ images/ codes/ tables/ flowcharts, used for GRPV system design or related to any other part of the report should be duly included in appendices.

Declaration

We, the undersigned students, understand that plagiarism is defined as anyone or a combination of the following:

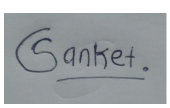
1. Uncredited verbatim copying of individual sentences, paragraphs, or illustrations (such as graphs, diagrams, etc.).

2. Uncredited improper paraphrasing of pages or paragraphs (by changing a few words or phrases or rearranging the original sentence order)

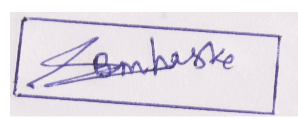
3. Credited verbatim copying a significant portion of a paper (or thesis chapter) without clearly delineating who did or wrote what.

I have ensured that all the ideas, expressions, graphs, diagrams, etc., that are not a result of my work are properly credited. Long phrases or sentences that had to be used verbatim from published literature have been identified using quotation marks.

I affirm that no portion of my work can be considered plagiarism, and I take full responsibility if such a complaint occurs. I understand fully well that the adviser/guide/co-guide of the seminar report may not be in a position to check for the possibility of such incidences of plagiarism in this body of work.



Sanket Bafana | 21d170037 | 18/04/2025



Sanket Mhaske | 21d170026 | 18/04/2025

Aditya Vibhute | 21d170003 | 18/04/2025

Pooja Mathary | 22b0358 | 18/04/2025

Declaration of individual contribution in GRPV Project

Group No: 05 Date: 18/04/2025

No	Criteria	Sanket Bafana 21d170037	Sanket Mhaske 21d170026	Aditya Vibhute 21d170003	Pooja Mathary 22b0358	Total
1	Communication & cooperation	25%	25%	25%	25%	**100%**
2	Quality of individual work	25%	25%	25%	25%	**100%**
3	Contribution to brainstorming & discussion	25%	25%	25%	25%	**100%**
4	Presentation and Report preparation	25%	25%	25%	25%	**100%**
5	Presentation delivery and Q&A	25%	25%	25%	25%	**100%**
	I hereby declare that I have reviewed and accepted the individual contributions documented for all group members.					
Signature		Sanket.	Smhaske	Aditya	Pooja	